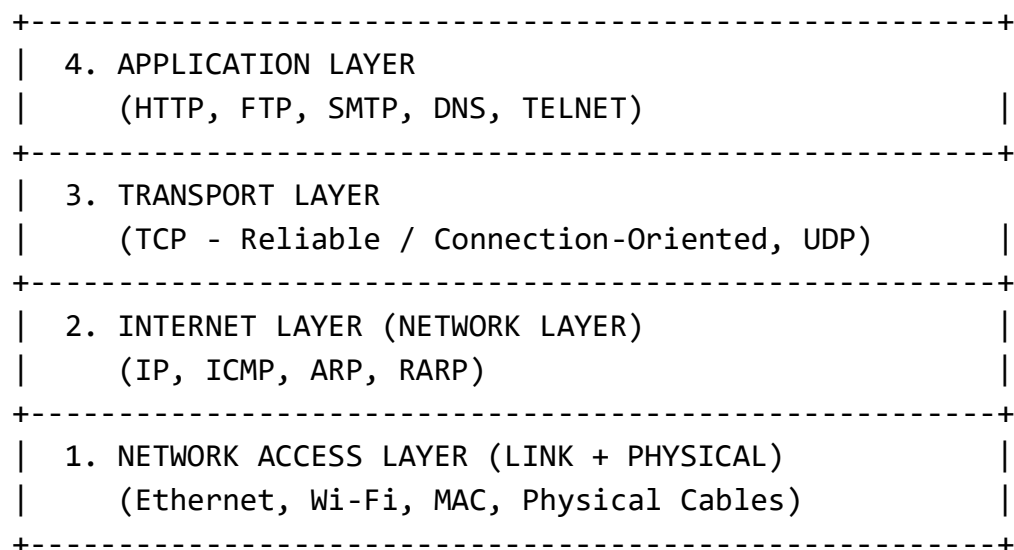


QUESTION 1

Q.1 (1) Draw the diagram of TCP/IP suite and Explain the function of Each Layer. (7 Marks)

1. Concept & TCP/IP Model Diagram:

TCP/IP (Transmission Control Protocol / Internet Protocol) model ek 4-layer functional architecture chhe je internet communications mate globally vapray chhe.



2. Functions of Each Layer:

- **1. Network Access Layer (Host-to-Network Layer):**
 - **Function:** OSI model na Physical ane Data Link Layer banne nu combination chhe. Aa layer physical medium (cables/radio waves) par actual bits ane frames transfer karva mate jababdar chhe.
 - **Key Tasks:** Physical (MAC) addressing, data framing, ane electrical/optical signaling.
 - **Protocols/Standards:** Ethernet (IEEE 802.3), Wi-Fi (IEEE 802.11).
- **2. Internet Layer:**
 - **Function:** Logical addressing ane routing dwara source host mathi packets ne destination host sudhi pahochadva nu kaam kare chhe.

- **Key Tasks:** Logical IP addressing provide kare chhe ane best optimal route select kare chhe. Connectionless datagram delivery service provide kare chhe.
- **Key Protocols:** IP (IPv4, IPv6), ICMP (Error reporting), ARP (IP to MAC translation).
- **3. Transport Layer:**
 - **Function:** Source ane destination application processes vachche end-to-end communication provide kare chhe.
 - **Key Tasks:** Port addressing (segmentation), multiplexing, flow control, ane error control.
 - **Key Protocols:**
 - **TCP (Transmission Control Protocol):** Connection-oriented, reliable, acknowledgement-based protocol.
 - **UDP (User Datagram Protocol):** Connectionless, unreliable, high-speed protocol.
- **4. Application Layer:**
 - **Function:** User application programs sathe directly interact kare chhe ane network services access karvani suvidha aape chhe.
 - **Key Protocols:** HTTP/HTTPS (Web browsing), FTP (File transfer), SMTP (Email delivery), DNS (Domain name resolution).

OR Q.1 (1) Draw and Explain the OSI Model for Computer Network. (7 Marks)

1. Concept & 7-Layer Diagram:

OSI (Open Systems Interconnection) model ISO dwara banavel ek 7-layer conceptual framework chhe je network systems na communication ne standardize kare chhe.

+-----+ Data Unit		
7. APPLICATION LAYER	(Network Service)	Data
6. PRESENTATION LAYER	(Format/Encrypt)	Data
5. SESSION LAYER	(Dialog Control)	Data
+-----+		
4. TRANSPORT LAYER	(End-to-End Conn)	Segments
+-----+		
3. NETWORK LAYER	(Path / IP Addr)	Packets
+-----+		
2. DATA LINK LAYER	(Hop-to-Hop/MAC)	Frames

+-----+		
	1. PHYSICAL LAYER (Physical Bits)	Bits
+-----+		

2. Functions of Each Layer (Bottom to Top):

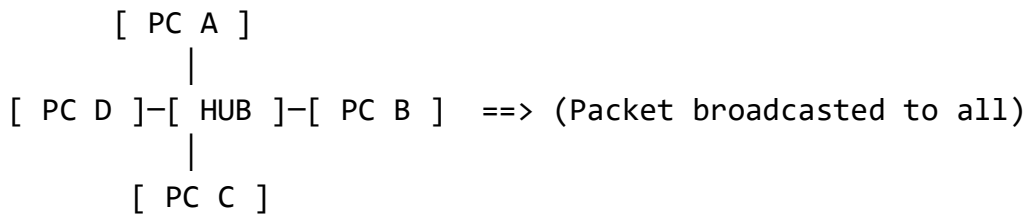
- **1. Physical Layer (Layer 1):**
 - Raw individual bits (0s and 1s) ne physical transmission media par transmit kare chhe.
 - Pin layouts, voltage levels, data rate (bit rate), ane transmission modes (simplex/duplex) define kare chhe.
- **2. Data Link Layer (Layer 2):**
 - Raw physical transmission ne error-free link ma convert kare chhe (Hop-to-Hop delivery).
 - **Key Functions:** Framing, Physical (MAC) Addressing, Flow Control, Error Detection (CRC).
- **3. Network Layer (Layer 3):**
 - Source thi Destination sudhi packets nu delivery ane routing kare chhe.
 - **Key Functions:** Logical IP Addressing, Routing table management, ane packet forwarding.
- **4. Transport Layer (Layer 4):**
 - Source host ni application thi Destination application sudhi End-to-End (Process-to-Process) reliable data delivery ensure kare chhe.
 - **Key Functions:** Service-point (Port) addressing, Segmentation & Reassembly, Connection Control.
- **5. Session Layer (Layer 5):**
 - Communicating devices vachche sessions (dialogs) establish, maintain, synchronize, ane terminate kare chhe.
 - Synchronization points (Checkpoints) add kare chhe jethi connection interrupt thay to pachhe thi restore kari shakay.
- **6. Presentation Layer (Layer 6):**
 - Data formatting, syntax, ane semantics sathe deal kare chhe.
 - **Key Functions:** Translation (ASCII to EBCDIC), Data Encryption/Decryption, ane Data Compression.
- **7. Application Layer (Layer 7):**
 - Direct user interface provide kare chhe ane network applications ne services provide kare chhe (e.g., Browsing, Email).

Q.1 (2) Compare Hub, Switch and Router. (7 Marks)

Parameter	Hub	Switch	Router
OSI Layer	Physical Layer (Layer 1)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Addressing Used	Koi address use kartu nathi (Non-intelligent)	Physical Address (MAC Address)	Logical Address (IP Address)
Data Unit	Electrical/Optical Bits	Frames	Packets
Data Transmission	Aavela data ne badha ports par Broadcast kare chhe	Specific target MAC address par Unicast kare chhe	Best route select kari packets ne Route/Forward kare chhe
Collision Domain	Ek j single/large collision domain hoy chhe	Darek individual port ek alag collision domain hoy chhe	Darek interface alag collision domain hoy chhe
Broadcast Domain	Single broadcast domain	Single broadcast domain (broadcast pass kare chhe)	Broadcast domain ne Break/Isolate kare chhe
Routing / Tables	Koi table maintain kartu nathi	MAC Address Table (CAM Table) banave chhe	Routing Table banave chhe
Network Type	Same LAN segment na nodes jode	Same LAN na devices interconnect kare	Alag-alag networks (LAN to WAN / Internet) jode

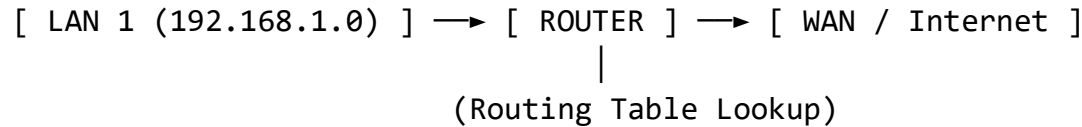
OR Q.1 (2) Write short note on Hub and Router. (7 Marks)**A. Hub:**

- **Concept:** Hub ek non-intelligent networking hardware device chhe je **OSI Physical Layer (Layer 1)** par kaam kare chhe. Aane **Multi-port Repeater** pan kahevay chhe.
- **Working:** Star topology ma center point tarike work kare chhe. Jyare koi ek node mathi data aave chhe, tyare te address check karya vagar baki badha ports par **broadcast** kari de chhe.
- **Types:**
 - *Passive Hub:* Sirf wire connect kare chhe, power jaruri nathi, signal amplify kartu nathi.
 - *Active Hub:* Signals ne regenerate ane amplify kare chhe (external power joie).
 - *Intelligent Hub:* Traffic monitor ane manage karvani feature provide kare chhe.
- **Limitation:** Network congestion vadhare chhe ane Half-Duplex mode par chale chhe.



B. Router:

- **Concept:** Router ek intelligent networking device chhe je **OSI Network Layer (Layer 3)** par kaam kare chhe. Te be ke tethi vadhare alag-alag networks (Local LAN to Public Internet/WAN) ne interconnected kare chhe.
- **Working:** Incoming packets na header mathi **Destination IP Address** read kare chhe, potana **Routing Table** sathe match kare chhe, ane packet ne best available path par forward kare chhe.
- **Key Features:**
 - Broadcast traffic ne forward kartu nathi, etle te **broadcast domain ne divide/isolate** kare chhe.
 - **Routing Support:** Static Routing (Manual entries) ane Dynamic Routing (RIP, OSPF protocols) bane support kare chhe.
 - Security provide kare chhe ane NAT (Network Address Translation) enable kare chhe.



QUESTION 2

Q.2 (1) Write Short Note on LAN and MAN. (5 Marks)

1. Local Area Network (LAN):

- **Geographical Span:** Ek small geographical area jem ke ek single room, office building, school ke college campus (1 km sudhi) ma limited hoy chhe.
- **Ownership:** Private ownership hoy chhe (single organization/person manage kare chhe).
- **Speed & Error Rate:** High data transfer speed (100 Mbps thi 10 Gbps) ane low transmission error rate hoy chhe.
- **Technology Used:** Ethernet cables (Twisted pair), Wi-Fi, Switches, Hubs.

2. Metropolitan Area Network (MAN):

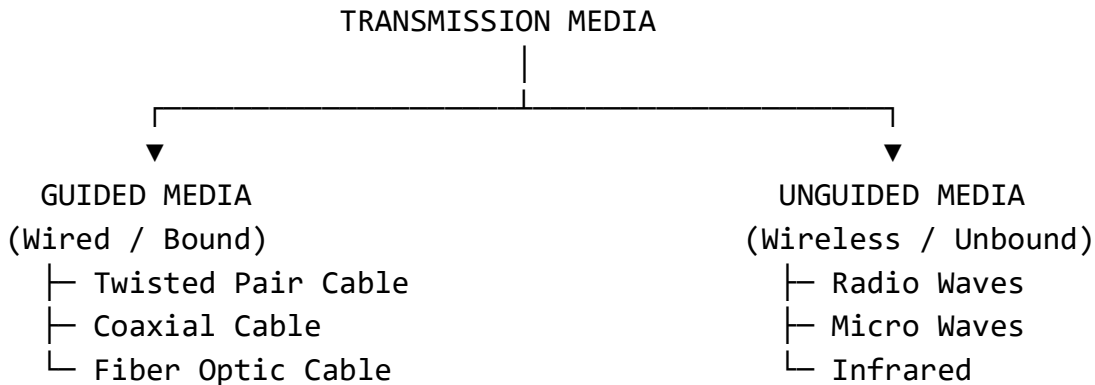
- **Geographical Span:** Aakha city (shehar) ke large town ne cover kare chhe (typically 5 km thi 50 km).
- **Ownership:** Private organization ke Public utility provider (e.g., Cable TV operator, Telecom company) dwara owned hoy chhe.
- **Speed & Error Rate:** Moderate speed (10 Mbps thi 1 Gbps) ane LAN karta thodo vadhare error rate hoy chhe.
- **Technology Used:** Fiber Optic cables, Optical networks, Coaxial lines.

Quick Comparison:

- **Area:** LAN = Single building; MAN = Entire city.
- **Cost:** LAN = Low design/installation cost; MAN = Higher infrastructure cost.
- **Example:** College computer lab network (LAN); City cable television network ke city-wide Wi-Fi (MAN).

Q.2 (2) Explain Types of Different Transmission Media. (5 Marks)

Transmission media physical communication path chhe jena dwara data signals sender thi receiver sudhi travel kare chhe. Tene 2 mukhya types ma classify karva ma aave chhe:



1. Guided Media (Wired / Bounded Media):

Physical solid conductors dwara signals transmit thay chhe.

- **Twisted Pair Cable:**
 - Be insulated copper conductors ek-beeja sathe spiral twist thayela hoy chhe jethi electromagnetic interference (cross-talk) ochhu thay.
 - *Types:* UTP (Unshielded Twisted Pair - e.g., Cat 5e/Cat 6) ane STP (Shielded Twisted Pair). LAN networks ma widely vapray chhe.
- **Coaxial Cable:**
 - Central copper conductor hoy chhe, jeni aaspas dielectric insulation, braided metal shielding, ane outer plastic jacket hoy chhe. High frequency signals ane Cable TV network mate upyogi chhe.
- **Fiber Optic Cable:**
 - Glass ke plastic core mathi data ne **Light Pulses** tarike transmit kare chhe.
 - Principle: **Total Internal Reflection (TIR)**.
 - Features: Highest bandwidth speed, no EMI interference, long-distance communication.

2. Unguided Media (Wireless / Unbounded Media):

Koi physical wire vagar electromagnetic radio waves no upyog space/air dwara transmission mate thay chhe.

- **Radio Waves:** Omnidirectional (badhi direction ma faelay chhe), walls penetrate kari shake chhe (e.g., AM/FM Radio, Wi-Fi).
- **Microwaves:** Unidirectional (Line-of-Sight), high-frequency point-to-point communication (e.g., Satellite link, Mobile towers).

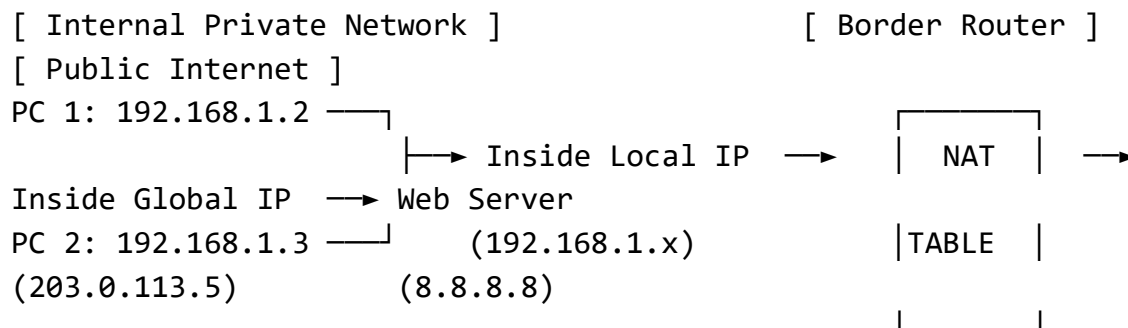
- **Infrared:** Short-range communication, solid obstacles/walls penetrate kari shaktu nathi (e.g., TV Remotes).

Q.2 (3) Draw and Explain the Function of Network Address Translation (NAT). (5 Marks)

1. Concept of NAT:

- NAT (Network Address Translation) ek routing method chhe je local private network na un-routable **Private IP Addresses** ne internet par chali shake teva valid **Public IP Address** ma translate kare chhe.
- IPv4 addresses ni achhat (exhaustion) ne rokvano aa sauthi best practical solution chhe.

2. NAT Diagram:



3. Working of NAT:

1. Private network no host (e.g., 192.168.1.2) internet na server mate request packet mokle chhe.
2. Packet jyare border router (NAT router) par pahonche chhe, tyare router private Source IP ne potana single public IP (203.0.113.5) sathe replace kare chhe.
3. Router aa mapping potana **NAT Translation Table** (IP + Port Number) ma record kare chhe.
4. Jyare internet mathi response packet aave chhe, router NAT table verify kari ne packet ne original private internal host tarf forward kari de chhe.

4. Key Functions / Advantages:

- **IPv4 Address Conservation:** Ek j single public IP no upyog kari multiple private devices internet access kari shake chhe.

- **Security:** Internal hosts na actual IP internet par hide rahe chhe, jethi direct outside cyber attacks thavanu jokham ghatse.

Q.2 (4) Differentiate IPv4 and IPv6. (5 Marks)

Parameter	IPv4 (Internet Protocol v4)	IPv6 (Internet Protocol v6)
Address Length	32 bits (\$4\$ bytes) long.	128 bits (\$16\$ bytes) long.
Total Address Space	$2^{32} \approx 4.3 \times 10^9$ (\$4.3 \text{ Billion}\$ addresses).	$2^{128} \approx 3.4 \times 10^{38}$ (Enormous addresses).
Representation Format	Dotted Decimal (e.g., 192.168.1.1).	Hexadecimal separated by colons (e.g., 2001:0db8::1).
Header Size	Variable length (\$20\$ to \$60\$ bytes).	Fixed base header length (\$40\$ bytes).
Configuration	Manual ke DHCP configured.	Stateless Auto-configuration (SLAAC) support.
Security (IPSec)	Optional (external protocol).	Built-in mandatory support.
Checksum in Header	Checksum field include hoy chhe.	Checksum field remove kareli chhe (faster processing).
Transmission Mode	Broadcast, Multicast, Unicast.	Anycast, Multicast, Unicast (No Broadcast).

QUESTION 3

Q.3 (1) Explain DHCP and ARP Protocol in Detail. (6 Marks)

A. DHCP (Dynamic Host Configuration Protocol):

- **Concept:** DHCP Application layer protocol chhe je network na clients (PC, Mobile) ne automatically IP configuration allocate kare chhe.
- **Key Parameters Provided:** Dynamic IP Address, Subnet Mask, Default Gateway, and DNS Server IP.
- **DORA Process (Working Mechanism):**
 - **D - Discover:** Client boot thai ne network ma broadcast kare chhe: *"Koi DHCP Server chhe?"*
 - **O - Offer:** DHCP Server client ne available IP offer kare chhe.
 - **R - Request:** Client offer accept kari IP address book karva request kare chhe.
 - **A - Acknowledge:** Server client ne confirmation (lease time sathe) acknowledgement mokle chhe.

Client ——— Discover (Broadcast) ———> Server
Client <—— Offer (Unicast/Bcast) ——— Server
Client ——— Request (Broadcast) ———> Server
Client <—— Ack (Unicast/Bcast) ——— Server

B. ARP (Address Resolution Protocol):

- **Concept:** Network Layer protocol chhe je known Logical IP address par thi unknown Physical Hardware (MAC) address find karva mate vapray chhe.
- **Working Steps:**
 - **ARP Request:** Jyare host pase target no IP hoy pan MAC nathi hoto, tyare te aakha network ma **Broadcast** kare chhe: *"Who has IP 192.168.1.5? Send me your MAC!"*
 - **ARP Reply:** Matching IP dharavto host potana MAC address sathe sender ne **Unicast reply** mokle chhe.
 - **ARP Cache:** Host aavele result ne future lookup mate potana internal ARP cache table ma store kari le chhe.

OR Q.3 (1) Explain Classful IP Addressing System. (6 Marks)**1. Concept:**

IPv4 addressing system has 32-bit addresses and is divided into five classes (Class A to Class E). Each address is divided into two parts: **Network ID (NetID)** and **Host ID (HostID)**.

```

Class A: | 0 |      Network ID (7 bits) |      Host ID (24 bits)
|
Class B: | 10 |      Network ID (14 bits) |      Host ID (16 bits)
|
Class C: | 110 |      Network ID (21 bits) | Host ID (8
bits) |
Class D: | 1110 |      Multicast Groups (28 bits)
|
Class E: | 1111 |      Experimental / Research Use (28 bits)
|

```

2. Detailed Breakdown Table of Classes:

Class	First Octet Range	Format (Net / Host)	Default Subnet Mask	Maximum Networks / Hosts per Network	Purpose / Application
Class A	\$1 - 126\$ (\$0 is reserved)	Net.Host.Host.Host	255.0.0.0	\$2^{24}\$ Networks / \$2^{24}\$ Hosts	Very Large Organizations / ISPs
Class B	\$128 - 191\$	Net.Net.Host.Host	255.255.0.0	\$2^{16}\$ Networks / \$2^{16}\$ Hosts	Medium to Large Universities

					es / Enterprise
Class C	\$192 - 223\$	Net.Net.Net.Host	255.255.255.0	$\approx 2 \times 10^6$ Million Nets) / 254×10^8 Hosts)	Small LANs / Small Businesses
Class D	\$224 - 239\$	No NetID / HostID	N/A	Reserved for Multicast	Audio/Video Multicast Streaming
Class E	\$240 - 255\$	No NetID / HostID	N/A	Reserved	R&D, Experimental testing